

Python datetime module

```
# datetime: is a built-in python module used to work with:
# - dates
# - time
# - timestamps
# - Date/Time calculations
# - Formatting and parsing dates

# import the module, and this contains several important classes
# - date
# - time
# - timedelta
# - timezone
from datetime import datetime, date, time, timedelta, timezone

# -----
# -- for example --
# -----
# date
# 2026-09-07

# time
# 05:19:30

# datetime
# 2026-09-07 05:19:30

# timedelta
# 3 days, 2 hours

# get the current date and time
now = datetime.now()
print(now)      # 2026-09-06 17:23:56.952109

# -----
# -- extract individual components --
# -----

# get the year
print(now.year)    # 2026

# get the month
print(now.month)   # 9

# print the day
print(now.day)     # 6

# print the hour
print(now.hour)    # 17

# print the minute
print(now.minute)  # 9

# print the second
print(now.second)  # 18

# -----
# -- creating a specific datetime --
# -----
# datetime(year, month, day, hour, minute, second)

# create specific date & time
dt = datetime(2026, 9, 7, 10, 30, 0)
print(dt)          # 2026-09-07 10:30:00
```

```

dt = datetime(2026, 1, 15)
print(dt)          # 2026-01-15 00:00:00

# get the current date only without time
today = date.today()
print(today)       # 2026-09-06

# get the year, month, day on the current date
print(today.year)  # 2026
print(today.month) # 9
print(today.day)   # 6

# =====
# == timedelta ==
# =====
# - timedelta: represents a duration or difference between two dates/times

# adding time
now = datetime.now()
print(now)          # 2026-09-06 17:36:34.538854

future = now + timedelta(days=10)
print(future)       # 2026-09-16 17:36:34.538854

dt = datetime(2026, 9, 7, 10, 0, 0)
print(dt)           # 2026-09-07 10:00:00

new_dt = dt + timedelta(days=2)
print(new_dt)       # 2026-09-09 10:00:00

# subtract time
dt = datetime(2026, 9, 7, 10, 0, 0)

new_dt = dt - timedelta(hours=5)
print(new_dt)       # 2026-09-07 05:00:00

# check timestamp from the last 10 minutes
now = datetime.now()
ten_minutes_ago = now - timedelta(minutes=10)
print(ten_minutes_ago) # 2026-09-06 18:42:31.673541

# =====
# == comparing date and time ==
# =====
dt1 = datetime(2026, 9, 7, 10, 0)
dt2 = datetime(2026, 9, 7, 12, 0)

print(dt1 < dt2)    # True
print(dt1 > dt2)    # False
print(dt1 == dt2)   # False

# strftime(): format datetime as a string
# - datetime -> string

dt = datetime(2026, 9, 7, 10, 30, 45)
formatted = dt.strftime("%Y-%m-%d")
print(formatted)    # 2026-09-07

```

```
formatted = dt.strftime("%Y-%m-%d %H:%M:%S")
print(formatted)          # 2026-09-07 10:30:45
```

Code	Meaning
%Y	4-digit year
%m	month
%d	day
%H	hour
%M	minute
%S	second

```
filename = datetime.now().strftime("%Y-%m-%d")
print(f'filename: {filename}')    # filename: 2026-09-06
```

```
# strptime(): convert a string into a datetime object
# - string -> datetime
```

```
timestamp = "2026-09-07 10:30:45"
```

```
dt = datetime.strptime(
    timestamp,
    "%Y-%m-%d %H:%M:%S"
)
```

```
print(dt)          # 2026-09-07 10:30:45
print(type(dt))    # <class 'datetime.datetime'>
```

```
dt = datetime.strptime(
    "07/09/2026 10:30:45",
    "%d/%m/%Y %H:%M:%S"
)
print(dt)          # 2026-09-07 10:30:45
```

```
# timezone: this is slightly more advanced but important
# there are two types
# 1- naive datetime
# 2- timezone-aware datetime
```

```
# naive timezone
dt = datetime.now()
print(dt)          # 2026-09-06 19:02:42.148443
```

```
# timezone-aware datetime
dt = datetime.now(timezone.utc)
print(dt)          # 2026-09-06 16:03:37.835429+00:00
```

```
# you can create a specific UTC datetime
```

```
dt = datetime(
    2026,
    9,
    7,
    10,
    30,
    tzinfo=timezone.utc
)
print(dt)          # 2026-09-07 10:30:00+00:00
```